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0.1 Exercise

Exercise 0.1.1 ► Calculate

Let V be a vector space of finite dimension with basis $(e_1, \dots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \dots, e_n^*)$. Please prove in two different ways that for every $(\alpha, \beta) \in \Lambda^p V^* \times \Lambda^q V^*$,

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

One way is to use definition, the other is to calculate in basis using the next exercise.

Proof. 1.

$$\alpha \wedge \beta = \frac{(p+q)!}{p!q!} \text{Alt}(\alpha \otimes \beta)$$

$$\beta \wedge \alpha = \frac{(p+q)!}{p!q!} \text{Alt}(\beta \otimes \alpha)$$

Only need to show that

$$\text{Alt}(\alpha \otimes \beta) = (-1)^{pq} (\beta \otimes \alpha)$$

Where

$$\text{Alt}(\alpha \otimes \beta) = \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) \sigma(\alpha \otimes \beta)$$

Let $\tau = (p + 1, \dots, p + q, 1, 2, \dots, p)$, then $\text{sgn}(\tau) = (-1)^{pq}$. We have

$$\begin{aligned}
 \text{Alt}(\alpha \otimes \beta) &= \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) \sigma \circ \tau (\beta \otimes \alpha) \\
 &= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) (\text{sgn} \tau) (\sigma \circ \tau) (\beta \otimes \alpha) \\
 &= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma' \in S_{p+q}} (\text{sgn} \sigma') \sigma' (\beta \otimes \alpha) \\
 &= (-1)^{pq} \text{Alt}(\beta \otimes \alpha)
 \end{aligned}$$

2. Suppose

$$\begin{aligned}
 \alpha &= \sum_{i_1 \dots i_p} a_{i_1 \dots i_p} e_{i_1}^* \wedge \dots \wedge e_{i_p}^* \\
 \beta &= \sum_{j_1 \dots j_q} b_{j_1 \dots j_q} e_{j_1}^* \wedge \dots \wedge e_{j_q}^*
 \end{aligned}$$

Then

$$\begin{aligned}
 \alpha \wedge \beta &= \sum_{i_1 \dots i_p} a_{i_1 \dots i_p} b_{j_1 \dots j_q} (e^*)_I \wedge (e^*)_J = \sum_{I,J} a_I b_J e_{IJ} \\
 \beta \wedge \alpha &= \sum_{I,J} a_I b_J e_{JI} = (-1)^{pq} \sum_{I,J} a_I b_J e_{IJ} = (-1)^{pq} \alpha \wedge \beta
 \end{aligned}$$

□

Exercise 0.1.2

Let V be a vector space of finite dimension and let V^* denotes the dual space. Prove that if $(\theta_1, \dots, \theta_r)$ is a family of r covectors in V^* which are **linearly dependent**, then the following wedge product:

$$\theta_1 \wedge \dots \wedge \theta_r = 0.$$

Proof. Since $\theta_1 \wedge \dots \wedge \theta_r$ is multi-linear and alternative, take any vector $v^1, \dots, v^r$, we have

$$\theta_1 \wedge \dots \wedge \theta_r (v^1, \dots, v^r) = \det(\theta_i(v^j))_{ij}$$

Since $\theta_1, \dots, \theta_r$ are linealy dependent , the row vectors of the matrix $(\theta_i(v^j))_{ij}$ are as well. Thus $\det(\theta_i(v^j))_{ij} = 0$, and $\theta_1 \wedge \dots \wedge \theta_r(v^1, \dots, v^r) = 0$. Since $v^1, \dots, v^r$ are arbitrary , $\theta_1 \wedge \dots \wedge \theta_r = 0$. $\square$

Exercise 0.1.3 ► Basis of Grassmann algebra

Let V be a vector space of finite dimension with basis $(e_1, \dots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \dots, e_n^*)$. Please prove that

$$(e_{i_1}^* \wedge \dots \wedge e_{i_k}^*)_{1 \leq i_1 < \dots < i_k \leq n}$$

forms a basis of $\Lambda^k V^*$.

Proof. For $\alpha \in \Lambda^k V^*$, and $I = (i_1, \dots, i_k)$. Define

$$\alpha_I := \alpha(e_{i_1}, \dots, e_{i_k})$$

If I has a reputed index , then $\alpha_I = 0$, and if $J = \sigma I$, $\alpha_J = (\text{sgn} \sigma) \alpha_I$. We have

$$\sum_I' \alpha_I e_I^*(e_{j_1}, \dots, e_{j_k}) = \sum_I' \alpha_I \delta_J^I = \alpha_J = \alpha(e_{j_1}, \dots, e_{j_k})$$

Thus $\alpha = \sum_I' \alpha_I e_I^*$. Which shows that $e_{i_1}^*, \dots, e_{i_k}^*, 1 \leq i_1 < \dots < i_k \leq n$ spans the $\Lambda^k V^*$. To show that they are linealy independent. Suppose that

$$\sum_I k_I e_I^* = 0$$

Both sides act on e_J , we have $k_J = 0$. $\square$

Exercise 0.1.4 ► Grassmann algebra versus Polynomial functions on V .

Let V be a vector space of finite dimension and let V^* denotes the dual space. A p-multilinear map α on V is **symmetric** if

$$\alpha(v_1, \dots, v_n) = \alpha(v_{\sigma(1)}, \dots, v_{\sigma(n)})$$

for all permutations σ . Such α defines a homogeneous polynomial $\tilde{\alpha}$

defined by

$$\tilde{\alpha}(v) = \alpha(v, \dots, v)$$

for all $v \in V$. Then show that for symmetric multilinear maps α, β on V of respective p, q , the product $\alpha \circ \beta$ defined by

$$\alpha \circ \beta(v_1, \dots, v_{p+q}) := \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

is valued into symmetric multilinear map on V of degree $p + q$ and that

$$\alpha \circ \beta(v, \dots, v) = \alpha(v, \dots, v) \beta(v, \dots, v).$$

Could you compare with $\Lambda^\bullet V^*, \wedge$?

Exercise 0.1.5 ► Basic important examples of smooth manifolds

Please prove that the following topological spaces have a smooth manifold structure :

- The Euclidean sphere $S^n \subset \mathbb{R}^{n+1}$.
- The matrix groups $GL_n(\mathbb{R}), SL_n(\mathbb{R}), O_n(\mathbb{R})$, these are Lie groups of matrices. Each time could you tell me what is the dimension of these groups.

Proof. • Define F :

$$F(x^1, \dots, x^{n+1}) = x_1^2 + \dots + x_{n+1}^2$$

Then

$$S^n = F^{-1}(1)$$

$$dF_x = \nabla F(x) = (2x_1, \dots, 2x_{n+1}) \neq 0, \forall x \in \mathbb{R}^{n+1}$$

By regular value theorem, $S^n \subseteq \mathbb{R}^{n+1}$ is a regular embedded submanifold of $\mathbb{R}^{n+1}$, which itself is a manifold.

- 1. $\text{Mat}(n, \mathbb{R}) \simeq \mathbb{R}^{n^2}$ 是 n^2 维流形.
- 2. $GL(n, \mathbb{R}) = \{M \in \text{Mat}(n, \mathbb{R}) : \det M \neq 0\}$ 作为 $\text{Mat}(n, \mathbb{R})$ 的一个开子集也是一个 n^2 维流形.

3. $SL(n, \mathbb{R}) = \{M \in \text{Mat}(n, \mathbb{R}) : \det M = 1\}$

$$\begin{aligned}\det : \text{Mat}(n, \mathbb{R}) &\rightarrow \mathbb{R} \\ M &\mapsto \det M\end{aligned}$$

$SL(n, \mathbb{R}) = \det^{-1}(1)$. 取一个 $M \in \text{Mat}(n, \mathbb{R})$, $\det M = 1$. 要证明 $(d(\det))_M : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$. 按分量求偏导, 得到

$$\frac{\partial}{\partial a_{ij}} (\det M) = A_{ij} (\text{代数余子式})$$

由于 $\det M = 1$. 至少有一个 $A_{ij} \neq 0$. 从而 $(d(\det))_M$ 是满的.

4. 定义 $P : \text{Mat}(n, \mathbb{R}) \rightarrow \text{Sym}(n, \mathbb{R})$ (对称矩阵) $\simeq \mathbb{R}^{\frac{n(n+1)}{2}}$.

$$P(A) := A^T A$$

则 $O(n, \mathbb{R}) = P^{-1}(0)$. 对于 $A \in O(n, \mathbb{R})$.

$$(dP)_A : \text{Mat}(n, \mathbb{R}) \rightarrow \text{Sym}(n, \mathbb{R})$$

与上例不同, 这里用分量的形式去计算 Jacobi 是看不清楚的, 不过由于 P 本身足够简单, 我们可以用微分最原始的含义来观察. 考虑

$$P(A + \varepsilon B) - P(A) = \varepsilon (dP)_A(B) + O(\dots)$$

带入得到

$$(A^T + \varepsilon B^T)(A + \varepsilon B) - A^T A = \varepsilon (B^T A + A^T B) + O(\varepsilon)$$

于是 $(dP)_A(B) = B^T A + A^T B$. 只需要证明这个关于 B 的映射是满的. 为此, 解方程

$$A^T B + B^T A = C$$

取 $B = \frac{1}{2}AC$, 利用 A, C 对称, 带入得到

$$\frac{1}{2}A^T A C + \frac{1}{2}C^T A^T A = \frac{1}{2}(C + C^T) = C$$

故 $O(n, \mathbb{R})$ 是 $\text{Mat}(n, \mathbb{R})$ 的一个 $\frac{n(n-1)}{2}$ 维正则子流形. 注意到对于 $A \in O(n, \mathbb{R})$, $\det A = 1$ 或 $\det A = -1$, 可以窥见事实上 $O(n, \mathbb{R})$ 有两个连通分支.

□

Exercise 0.1.6 ► Quotient manifolds

Prove that the following objects are quotient manifolds

- $\mathbb{R}P^n$ obtained by quotienting S^n by antipodal map, show that it is nonorientable by a picture. Could you explain by a picture that the neighborhood of the line at infinity is homeomorphic to a Möbius ribbon ?
- Prove that the torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is a smooth manifold.

Proof.

- 1. See $\mathbb{R}P^n$ is $S^2/(x \sim -x)$. The orientation changes at the equator. The map $A : S^2 \rightarrow S^2, A(x) = -x$ induces an identity map on $\mathbb{R}P^2$, dA send an orientation basis $\{e_1, e_2\}$ to one opposite to it.
- 2. Use $\mathbb{R}P^2 = D^2/\{x \sim -x \text{ for } x \in \partial D^2 = S^1\}$. Then the neighborhood of the line at infinity is a thin ring near ∂D^2 with some quotient relation. The quotient is just the same as that of Möbius.

□

Exercise 0.1.7 ► Vector fields are derivation

Let M be a C^∞ manifold. Show that the left C^∞ module of derivation of the ring $C^\infty(M)$ i.e. linear maps $D : C^\infty(M) \rightarrow C^\infty(M)$ satisfying

$$D(fg) = (Df)g + f(Dg)$$

can be identified with vector fields on M , smooth sections of TM .

Proof. Define

$$D(x)f = (Df)(x)$$

Then

$$D(x)(fg) =$$

□

Smooth Manifold

1.1 Topological Manifolds

Definition 1.1.1

Suppose M is a topological space. We say that M is a **topological manifold of dimension n** or a **topological n -manifold** if it has the following properties:

- M is a **Hausdorff space** (T_2): for every pair of distinct points $p, q \in M$, there are disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
- M is **second-countable**: there exists a countable basis for the topology of M .
- M is **locally Euclidean of dimension n** : each point of M has a neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

Remark.

The third property means, more specifically, that for each $p \in M$ we can find

- an open subset $U \subseteq M$ containing p ,
- an open subset $\hat{U} \subseteq \mathbb{R}^n$, and
- a homeomorphism $\varphi : U \rightarrow \hat{U}$.

Remark.

A topological manifold M is locally compact, locally path-connected, separable, Lind  lof, σ -finite, T_3 , T_4 , and paracompact.

1.1.1 Coordinate Chart

Definition 1.1.2 ▶ Coordinate Chart

Let M be a topological n -manifold. A **coordinate chart** (or just a **chart**) on M is a pair (U, φ) , where U is an open subset of M and $\varphi : U \rightarrow \hat{U}$ is a homeomorphism from U to an open subset $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$.

Remark.

- By definition of a topological manifold, each point $p \in M$ is contained in the domain of some chart (U, φ) .
- If $\varphi(p) = 0$, we say that the chart is **centered at p** .
- If (U, φ) is any chart whose domain contains p , it is easy to obtain a new chart centered at p by subtracting the constant vector $\varphi(p)$.

1.2 Smooth Structures

Definition 1.2.1 ▶ Smoothly Compatible

Let M be a topological n -manifold.

- If $(U, \varphi), (V, \psi)$ are two charts such that $U \cap V \neq \emptyset$, the composite map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is called the **transition map from φ to ψ** .
- Two charts (U, φ) and (V, ψ) are said to be **smoothly compatible** if either $U \cap V = \emptyset$ or the transition map $\psi \circ \varphi^{-1}$ is a diffeomorphism.

Remark.

- $\psi \circ \varphi^{-1}$ is a composition of homeomorphisms, and is therefore itself a homeomorphism.
- Since $\varphi(U \cap V)$ and $\psi(U \cap V)$ are open subsets of $\mathbb{R}^n$, smoothness of this map is to be interpreted in the ordinary sense of having continuous partial derivatives of all orders.

Definition 1.2.2 ▶ Atlas

We define an **atlas for M** to be a collection of charts whose domains cover M . An atlas $\mathcal{A}$ is called a **smooth atlas** if any two charts in $\mathcal{A}$ are smoothly compatible with each other.

Definition 1.2.3 ► Maximal Atlas

A smooth atlas $\mathcal{A}$ on M is **maximal** if it is not properly contained in any larger smooth atlas. This just means that any chart that is smoothly compatible with every chart in $\mathcal{A}$ is already in $\mathcal{A}$.

Definition 1.2.4 ► Smooth Structures and Smooth Manifolds

- If M is a topological manifold, a **smooth structure on M** is a maximal smooth atlas.
- A **smooth manifold** is a pair $(M, \mathcal{A})$, where M is a topological manifold and $\mathcal{A}$ is a smooth structure on M .

Lemma 1.2.5 ► Smooth Manifold Chart Lemma

Let M be a set, and suppose we are given a collection $\{U_\alpha\}$ of subsets of M together with maps $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$, such that the following properties are satisfied:

1. For each α , φ_α is a bijection between U_α and an open subset $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.
2. For each α and β , the sets $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$.
3. Whenever $U_\alpha \cap U_\beta \neq \emptyset$, the map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth.
4. Countably many of the sets U_α cover M .
5. Whenever p, q are distinct points in M , either there exists some U_α containing both p and q or there exist disjoint sets U_α, U_β with $p \in U_\alpha$ and $q \in U_\beta$.

Then M has a unique smooth manifold structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

1.3 Manifolds with Boundary

Definition 1.3.1 ► Manifolds with Boundary

An **n -dimensional topological manifold with boundary** is a second-countable Hausdorff space M in which every point has a neighborhood homeomorphic either to an open subset of $\mathbb{R}^n$ or to a (relatively) open subset of $\mathbb{H}^n$.

Definition 1.3.2 ► Chart for Manifolds with Boundary

Let M be an n -dimensional topological manifold with boundary. An open subset $U \subseteq M$ together with a map $\varphi : U \rightarrow \mathbb{R}^n$ that is a homeomorphism onto an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$ will be called a **chart for M** .

- When it is necessary to make the distinction, we will call (U, φ) an **interior chart** if $\varphi(U)$ is an open subset of $\mathbb{R}^n$ (which includes the case of an open subset of $\mathbb{H}^n$ that does not intersect $\partial\mathbb{H}^n$).
- We call (U, φ) a **boundary chart** if $\varphi(U)$ is an open subset of $\mathbb{H}^n$ such that $\varphi(U) \cap \partial\mathbb{H}^n \neq \emptyset$.
- A boundary chart whose image is a set of the form $B_r(x) \cap \mathbb{H}^n$ for some $x \in \partial\mathbb{H}^n$ and $r > 0$ is called a **coordinate half-ball**.

Smooth Maps

2.1 Smooth Functions and Smooth Maps

2.1.1 Smooth Maps Between Manifolds

Definition 2.1.1 ▶ Smooth Maps Between Manifolds

Let M, N be smooth manifolds, and let $F : M \rightarrow N$ be any map. We say that F is a **smooth map** if for every $p \in M$, there exist smooth charts (U, φ) containing p and (V, ψ) containing $F(p)$ such that $F(U) \subseteq V$ and the composite map $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\psi(V)$.

Remark.

- By taking $N = V = \mathbb{R}^k$ and $\psi = \text{Id} : \mathbb{R}^k \rightarrow \mathbb{R}^k$, we get the definition of smoothness of real-valued or vector-valued functions.

2.2 Partitions of Unity

Definition 2.2.1 ▶ Support

If f is any real-valued or vector-valued function on a topological space M , the **support of f** , denoted by $\text{supp } f$, is the closure of the set of points where f is nonzero:

$$\text{supp } f = \overline{\{p \in M : f(p) \neq 0\}}.$$

- If $\text{supp } f$ is contained in some set $U \subseteq M$, we say that f is **supported in U** .
- A function f is said to be **compactly supported** if $\text{supp } f$ is a compact set, denoted by $f \in C_0^\infty(M)$.

Remark.

Clearly, every function on a compact space is compactly supported.

Lemma 2.2.2

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0, \end{cases}$$

is smooth.

Lemma 2.2.3

Given any real numbers r_1 and r_2 such that $r_1 < r_2$, there exists a smooth function $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $h(t) \equiv 1$ for $t \leq r_1$, $0 < h(t) < 1$ for $r_1 < t < r_2$, and $h(t) \equiv 0$ for $t \geq r_2$.

Proof. Let f be the function of the previous lemma, and set

$$h(t) = \frac{f(r_2 - t)}{f(r_2 - t) + f(t - r_1)}.$$

Note that the denominator is positive for all t , because at least one of the expressions $r_2 - t$ and $t - r_1$ is always positive. The desired properties of h follow easily from those of f . $\square$

Lemma 2.2.4 ► Bump Function on $\mathbb{R}^n$

Given any positive real numbers $r_1 < r_2$, there is a smooth function $H : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $H \equiv 1$ on $\bar{B}_{r_1}(0)$, $0 < H(x) < 1$ for all $x \in B_{r_2}(0) \setminus \bar{B}_{r_1}(0)$, and $H \equiv 0$ on $\mathbb{R}^n \setminus B_{r_2}(0)$.

Proof. Just set $H(x) = h(|x|)$, where h is the function of the preceding lemma. Clearly, H is smooth on $\mathbb{R}^n \setminus \{0\}$, because it is a composition of smooth functions there. Since it is identically equal to 1 on $B_{r_1}(0)$, it is smooth there too. $\square$

Theorem 2.2.5 ► Existence of Bump Function on Manifold

Let M be a smooth manifold, $K \subseteq M$ a compact set. $U \subseteq M$ is a open set such that $K \subseteq U$. Then there exists $\varphi \in C_0^\infty(M)$, such that $\text{supp } \varphi \subseteq U$, $0 \leq \varphi \leq 1$ and $\varphi \equiv 1$ on K .

Proof. For each $x \in K$, take a chart (φ_x, U_x, V_x) centering x . We may assume that $V_x \subseteq B_3(0)$ and $U_x \subseteq U$ by contracting when necessary. Let $\tilde{U}_x := \varphi_x^{-1}(B_1(0))$. We define

$$f_x(p) := \begin{cases} H \circ \varphi_x(p), & p \in U_x \\ 0, & p \notin U_x \end{cases}$$

Where H is defined in Lemma 2.2.4 and we take $r_1 = 1, r_2 = 2$. It is straightforward to verify that $f_x \equiv 1$ on $\tilde{U}_x$ and $f_x \equiv 0$ on U_x^c , $\text{supp } f_x = \varphi_x^{-1}(\overline{B_2(0)}) \subseteq U_x$. Since the homeomorphism keeps the closure $\varphi_x^{-1}(\overline{B_2(0)}) = \overline{\varphi_x^{-1}(B_2(0))}$.

Now K has a open cover $\{\tilde{U}_x\}$. Since K is compact, there are finitely many $\tilde{U}_{x_i}$ such that

$$\bigcup_{i=1}^N \tilde{U}_{x_i} \supseteq K$$

We define

$$\psi := \sum_{i=1}^N f_{x_i}$$

Then $\psi \geq 1$ on K , $\text{supp } \psi \subseteq U$.

Let $\rho : \mathbb{R} \rightarrow [0, 1]$ be a smooth function such that

$$\rho(t) = \begin{cases} 0, & t \leq 1/2 \\ 1, & t \geq 1 \end{cases}$$

and ρ is monotonically increasing on $(1/2, 1)$. Now, we define our final function $\varphi : M \rightarrow \mathbb{R}$ as the composition

$$\varphi := \rho \circ \psi$$

□

Definition 2.2.6 ▶ Partitions of Unity

Suppose M is a topological space, and let $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ be an arbitrary open cover of M , indexed by a set A . A **partition of unity subordinate to $\mathcal{X}$** is an indexed family $(\psi_\alpha)_{\alpha \in A}$ of continuous functions $\psi_\alpha : M \rightarrow \mathbb{R}$ with the following properties:

1. $0 \leq \psi_\alpha(x) \leq 1$ for all $\alpha \in A$ and all $x \in M$.
2. $\text{supp } \psi_\alpha \subseteq X_\alpha$ for each $\alpha \in A$.
3. The family of supports $(\text{supp } \psi_\alpha)_{\alpha \in A}$ is locally finite, meaning that every point has a neighborhood that intersects $\text{supp } \psi_\alpha$ for only finitely many values of α .
4. $\sum_{\alpha \in A} \psi_\alpha(x) = 1$ for all $x \in M$.

If M is a smooth manifold with or without boundary, a **smooth partition of unity** is one for which each of the functions ψ_α is smooth.

Remark.

- Because of the local finiteness condition (3), the sum in (4) actually has only finitely many nonzero terms in a neighborhood of each point, so there is no issue of convergence.
- For a manifold M , M is Lindelöf Space, we can always assume that $(X_\alpha)_{\alpha \in A}$ is a countable collection, thus the $(\psi_\alpha)_{\alpha \in A}$ is as well.

Theorem 2.2.7 ► Existence of Partitions of Unity

Suppose M is a smooth manifold with or without boundary, and $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ is any indexed open cover of M . Then there exists a smooth partition of unity subordinate to $\mathcal{X}$.

Applications

Proposition 2.2.8 ► Bump function Support on a Closed Set

Let M be a manifold, $A \subseteq M$ be a closed set, $U \supseteq A$ is an open set. Then there exists $\varphi \in C_0^\infty(M)$ such that $0 \leq \varphi \leq 1$, and $\varphi \equiv 1$ on A .

Proof. $\{U, M \setminus A\}$ is a open cover of M . Then there exists a partition of unity $\{\rho_1, \rho_2\}$ subordinate to it. Then ρ_1 is just the function we need. $\square$

Theorem 2.2.9 ► Stone-Weierstrass

Let X be a compact Hausdorff Space. $\mathcal{A} \subseteq C(X, \mathbb{R})$ is a subalgebra. Then $\mathcal{A}$ is dense in $C(X, \mathbb{R})$, if and only if it satisfies

1. Nowhere vanishing: For each $x \in X$, there exists $g \in \mathcal{A}$ such that $g(x) \neq 0$.
2. Separates points: For distinct $x, y \in X$, there exists $f \in \mathcal{A}$ such that

$$f(x) \neq g(x).$$

Theorem 2.2.10 ► Whitney

Let M be a smooth manifold. For each continuous function $g : M \rightarrow \mathbb{R}$, $\delta : M \rightarrow \mathbb{R}_+$, there exists a smooth function $f : M \rightarrow \mathbb{R}$, such that

$$|f(p) - g(p)| < \delta(p), \quad \forall p \in M$$

Theorem 2.2.11 ► Whitney Extension

Let M be a smooth manifold. $A \subseteq M$ is a closed set. For continuous $g : M \rightarrow \mathbb{R}$, $\delta : M \rightarrow \mathbb{R}_+$, where g is smooth on A . Then there exists smooth function $f : M \rightarrow \mathbb{R}$, such that

1.

$$f(p) = g(p), \quad \forall p \in A$$

2.

$$|f(p) - g(p)| < \delta(p), \quad \forall p \in M$$

Tangent Vectors

3.1 Tangent Space

3.1.1 Geometric Tangent Vectors

Definition 3.1.1

If a is a point of $\mathbb{R}^n$, a map $w : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$ is called a **derivation at a** if it is linear over $\mathbb{R}$ and satisfies the following product rule:

$$w(fg) = f(a)wg + g(a)wf.$$

Let $T_a\mathbb{R}^n$ denote the set of all derivations of $C^\infty(\mathbb{R}^n)$ at a . Clearly, $T_a\mathbb{R}^n$ is a vector space under the operations

$$(w_1 + w_2)f = w_1f + w_2f, \quad (cw)f = c(wf).$$

Proposition 3.1.2

Let $a \in \mathbb{R}^n$.

- (a) For each geometric tangent vector $v_a \in \mathbb{R}_a^n$, the map $D_v|_a : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$ defined by

$$D_v|_a f = D_v f(a) = \left. \frac{d}{dt} \right|_{t=0} f(a + tv).$$

is a derivation at a .

- (b) The map $v_a \mapsto D_v|_a$ is an isomorphism from $\mathbb{R}_a^n$ onto $T_a\mathbb{R}^n$.

3.1.2 Tangent Vectors on Manifolds

Definition 3.1.3

Let M be a smooth manifold with or without boundary, and let p be a point of M . A linear map $v : C^\infty(M) \rightarrow \mathbb{R}$ is called a **derivation at p** if

it satisfies

$$v(fg) = f(p)vg + g(p)vf \quad \text{for all } f, g \in C^\infty(M). \quad (3.1)$$

The set of all derivations of $C^\infty(M)$ at p , denoted by T_pM , is a vector space called the **tangent space to M at p** . An element of T_pM is called a **tangent vector at p** .

Lemma 3.1.4 ► Properties of Tangent Vectors on Manifolds

Suppose M is a smooth manifold with or without boundary, $p \in M$, $v \in T_pM$, and $f, g \in C^\infty(M)$.

- (a) If f is a constant function, then $vf = 0$.
- (b) If $f(p) = g(p) = 0$, then $v(fg) = 0$.

3.2 The Differential of a Smooth Map

3.2.1 Differential on Euclidean Space

Lemma 3.2.1

Let $U \subseteq \mathbb{R}^m, V \subseteq \mathbb{R}^n$ are open sets. $f : U \rightarrow V$ is smooth. Take $a \in U$, define a map

$$\begin{aligned} (df)_a : T_aU &\rightarrow T_{f(a)}V \\ (df_a)(D_v^a) &\mapsto D_w^{f(a)}, \quad w = J_f(a)v \end{aligned}$$

Then for each $g \in C^\infty(V)$, we have

$$((df_a)(v))(g) = D_v^a(g \circ f)$$

Note.

$(df_a)(v)$ 对应的导子恰好是在点 $f(a)$ 处将 $g \in C^\infty(V)$ 映到 $D_v^a(g \circ f)$ 的导子.

Definition 3.2.2

If M and N are smooth manifolds with or without boundary and $F :$

$M \rightarrow N$ is a smooth map, for each $p \in M$ we define a map

$$dF_p : T_p M \rightarrow T_{F(p)} N,$$

called the **differential of F at p** . Given $v \in T_p M$, we let $dF_p(v)$ be the derivation at $F(p)$ that acts on $f \in C^\infty(N)$ by the rule

$$dF_p(v)(f) = v(f \circ F).$$

Proposition 3.2.3 ► Properties of Differentials

Let M , N , and P be smooth manifolds with or without boundary, let $F : M \rightarrow N$ and $G : N \rightarrow P$ be smooth maps, and let $p \in M$.

- (a) $dF_p : T_p M \rightarrow T_{F(p)} N$ is linear.
- (b) $d(G \circ F)_p = dG_{F(p)} \circ dF_p : T_p M \rightarrow T_{G \circ F(p)} P$.
- (c) $d(\text{Id}_M)_p = \text{Id}_{T_p M} : T_p M \rightarrow T_p M$.
- (d) If F is a **diffeomorphism**, then $dF_p : T_p M \rightarrow T_{F(p)} N$ is an isomorphism, and $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.

3.3 Tangent Bundle

Definition 3.3.1

Given a smooth manifold M with or without boundary, we define the **tangent bundle of M** , denoted by TM , to be the disjoint union of the tangent spaces at all points of M :

$$TM = \bigsqcup_{p \in M} T_p M.$$

Remark.

- **projection map** $\pi : TM \rightarrow M$, $\pi(p, v) = p$.

Proposition 3.3.2

For any smooth n -manifold M , then tangent bundle TM has a natural topology and smooth structure that make it into a $2n$ -dimensional smooth manifold. With respect to this structure, the projection $\pi : TM \rightarrow M$ is smooth.

Proof. Given a chart (U, φ) for M , $\pi^{-1}(U) \subseteq TM$ is the set of all tangent vectors to M at all points of U . Define a map $\tilde{\varphi} : \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}$ by

$$\tilde{\varphi}\left(v^i \frac{\partial}{\partial x^i} \Big|_p\right) = (x^1(p), \dots, x^n(p), v^1, \dots, v^n)$$

□

Submersions, Immersions and Embeddings

4.0.1 Local Diffeomorphism

Definition 4.0.1

If M and N are smooth manifolds with or without boundary, a map $F : M \rightarrow N$ is called a **local diffeomorphism** if every point $p \in M$ has a neighborhood U such that $F(U)$ is open in N and $F|_U : U \rightarrow F(U)$ is a diffeomorphism.

Theorem 4.0.2 ► Inverse Function Theorem for Manifolds

Suppose M and N are smooth manifolds, and $F : M \rightarrow N$ is a smooth map. If $p \in M$ is a point such that dF_p is invertible, then there are connected neighborhoods U_0 of p and V_0 of $F(p)$ such that $F|_{U_0} : U_0 \rightarrow V_0$ is a diffeomorphism.

Note.

也就是说, 光滑映射在微分映射可逆的点附近是微分同胚.

Submanifolds

5.1 Embedded Submanifolds

5.1.1 Slice Charts for Embedded Submanifolds

Definition 5.1.1

If U is an open subset of $\mathbb{R}^n$ and $k \in \{0, \dots, n\}$, a **k -dimensional slice of U** (or simply a **k -slice**) is any subset of the form

$$S = \{(x^1, \dots, x^k, x^{k+1}, \dots, x^n) \in U : x^{k+1} = c^{k+1}, \dots, x^n = c^n\}$$

for some constants $c^{k+1}, \dots, c^n$.

Definition 5.1.2 ► local k -slice condition

Let M be a smooth n -manifold.

- Let (U, φ) be a smooth chart on M . If S is a subset of U such that $\varphi(S)$ is a k -slice of $\varphi(U)$, then we say that S is a **k -slice of U** .
- Given a subset $S \subseteq M$ and a nonnegative integer k , we say that S satisfies the **local k -slice condition** if each point of S is contained in the domain of a smooth chart (U, φ) for M such that $S \cap U$ is a single k -slice in U .
- Any such chart is called a **slice chart for S in M** , and the corresponding coordinates $(x^1, \dots, x^n)$ are called **slice coordinates**.

5.1.2 Level Sets

Definition 5.1.3

If $\Phi : M \rightarrow N$ is a smooth map.

- A point $p \in M$ is said to be a **regular point of Φ** if $d\Phi_p : T_p M \rightarrow T_{\Phi(p)} N$ is surjective; it is a **critical point of Φ** otherwise.
- A point $c \in N$ is said to be a **regular value of Φ** if every point of the level set $\Phi^{-1}(c)$ is a regular point, and a **critical value** otherwise.

Theorem 5.1.4 ► Constant-Rank Level Set Theorem

Let M and N be smooth manifolds, and let $\Phi : M \rightarrow N$ be a smooth map with constant rank r . Each level set of Φ is a properly embedded submanifold of codimension r in M .

Proof. Write $m = \dim M$, $n = \dim N$, and $k = m - r$. Let $c \in N$ be arbitrary, let S denote the level set $\Phi^{-1}(c) \subseteq M$. From the rank theorem, for each $p \in S$, there are smooth charts (U, φ) centered at p and (V, ψ) centered at $c = \Phi(p)$ in which Φ has a coordinate representation of the form

$$\psi \circ \Phi \circ \varphi^{-1}(x^1, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0)$$

and therefore $S \cap U$, then

$$\begin{aligned} q \in S &\iff \Phi(q) = c \iff (x^1(q), \dots, x^r(q), 0, \dots, 0) = (0, \dots, 0) \\ &\iff x^1(q) = \dots = x^r(q) = 0 \end{aligned}$$

Thus

$$S \cap U = \{(x^1, \dots, x^r, x^{r+1}, \dots, x^m) \in U : x^1 = \dots = x^r = 0\}$$

is the k -slice. So S satisfies the local k -slice condition, so it is an embedded submanifold of dimension k . $\square$

Sard's Theorem

6.1 Set of Measure Zero

Lemma 6.1.1

Suppose $A \subseteq \mathbb{R}^n$ is a compact subset whose intersection with $\{c\} \times \mathbb{R}^{n-1}$ has $(n-1)$ -dimensional measure zero for every $c \in \mathbb{R}$. Then A has a n -dimensional measure zero.

Proof sketch.

把 A 的每一个“片” $A_c := \{(c, x) \in \{c\} \times \mathbb{R}^{n-1} : (c, x) \in A\}$ 的开覆盖, 缩短成里面的一个开的“方片”. 这些“方片”构成 A 的一个开覆盖, 取出一个有限子覆盖计算其 Volume, 将“方片”的相交部分抹去, 使得总体积可控.

Proof. Take $[a, b] \subseteq \mathbb{R}$ such that $A \subseteq [a, b] \times \mathbb{R}^{n-1}$.

Let $A_c := \{(c, x) \in \{c\} \times \mathbb{R}^{n-1} : (c, x) \in A\}$. The projection from A to $[a, b] \times \mathbb{R}^{n-1}$ is continuous map from a compact set to a Hausdorff space, thus is a proper map. It follows that A_c is a compact subspace of A for all $c \in [a, b]$.

Take any $\delta > 0$. There exists a finite open cover $C_1, \dots, C_k$ of A_c , such that the total volume is less than δ . Denote the open subset $C_1 \cup \dots \cup C_k \subseteq A_c$ by U_c . We claim that there is an open interval J_c containing c and contained in U_c , such that A_c is contained in the intersection of A with $J_c \times \mathbb{R}^{n-1}$: otherwise, there is a sequence $\{(c_i, x_i)\}$, such that $c_i \rightarrow c$, $x_i \notin U_c$; but by passing to a subsequence $x_i \rightarrow A_c \setminus U_c$, which is a contradiction to $A_c \subseteq U_c$.

The intervals $\{J_c : c \in [a, b]\}$ form an open cover of $[a, b]$, so there is a finite many $c_1 < c_2 < \dots < c_m$ such that $J_{c_1}, \dots, J_{c_m}$ form a finite open cover of A . We may assume that the total length is less than $2|b-a|$ by shrinking the intersection of the members. Then $J_{c_1}, \dots, J_{c_m}$ is an open cover of A with total volume less than $2\delta|b-a|$, which can be arbitrarily small. Hence A has a n -dimensional measure zero. $\square$

Proposition 6.1.2

Suppose A is an open or closed subset of $\mathbb{R}^{n-1}$ or $\mathbb{H}^{n-1}$, and $f : A \rightarrow \mathbb{R}$ is a continuous function. Then the graph of f has measure zero in $\mathbb{R}^n$.

Proof sketch.

对于 A 是紧集的情况, 令 $P^n = \{\text{Sets are } n\text{-dimensional measure zero}\}$. 上面的引理给出了 $\forall c, P_c^{n-1} \implies P^n$ 的结论. 因此降维操作是无损的, 从而归纳法有效.

Proof. First assume that A is compact. We prove the theorem in this case by induction on n . When $n = 1$, it is trivial because the graph of f is at most a point. To prove the inductive step, we appeal to lemma above. For each $c \in \mathbb{R}$, the intersection of the graph of f with $\{c\} \times \mathbb{R}^{n-1}$ is just the graph of the restriction of f to $\{x \in A : x^1 = c\}$, which is in turn the graph of continuous functions of $(n-2)$ variables. It follows by induction that each such graph has $(n-1)$ -dimensional measure zero, and thus by lemma above, the graph of f itself has n -dimensional measure zero.

If A is noncompact, it is a countable union of compact subsets, so the graph of f is a countable union of sets of measure zero. $\square$

Corollary 6.1.3

Every proper affine subspace of $\mathbb{R}^n$ has measure zero in $\mathbb{R}^n$.

Proposition 6.1.4 ▶ Diffeomorphism-invariant of Measure Zero

Suppose $A \subseteq \mathbb{R}^n$ has measure zero and $F : A \rightarrow \mathbb{R}^n$ is a smooth map. Then $F(A)$ has measure zero.

Definition 6.1.5

If M is a smooth n -manifold with or without boundary, we say that a subset $A \subseteq M$ has **measure zero in M** if for every smooth chart (U, φ) for M , the subset $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has n -dimensional measure zero.

Remark.

Equivalently, $\forall \varepsilon > 0$, there are countably many charts (U_i, φ_i) , such that

$$A \subseteq \bigcup_{i=1}^{\infty} U_i, \quad \sum_{i=1}^{\infty} [(\varphi_i^{-1})_{\#} \mu](U_i) < \varepsilon$$

Lemma 6.1.6

Let M be a smooth n -manifold with or without boundary and $A \subseteq M$. Suppose that for some collection $\{(U_{\alpha}, \varphi_{\alpha})\}$ of smooth charts whose domains cover A , $\varphi_{\alpha}(A \cap U_{\alpha})$ has measure zero in $\mathbb{R}^n$ for each α . Then A has measure zero in M .

Proof. Let (V, ψ) be any smooth chart for M . We need to show that $\psi(A \cap V)$ has measure zero. There exists countably many U_{α} 's covering $A \cap V$. We write

$$\psi(A \cap V \cap U_{\alpha}) = (\psi \circ \varphi_{\alpha}^{-1}) \circ \varphi_{\alpha}(A \cap V \cap U_{\alpha})$$

$\varphi_{\alpha}(A \cap V \cap U_{\alpha})$ is a subset of measure zero set $\varphi_{\alpha}(A \cap U_{\alpha})$, thus has measure zero as well. It follows from Proposition 6.1.4 that $\psi(A \cap V \cap U_{\alpha})$ has measure zero. Then $\psi(A \cap V)$ is a countable unions of measure zero sets, thus is measure zero as well. $\square$

Exercise 6.1.7

Let M be a smooth manifold with or without boundary. Show that a countable union of sets of measure zero in M has measure zero.

Proof. Suppose $A_1, A_2, \dots$ are sets of measure zero in M . Let $A = \bigcup_{i=1}^{\infty} A_i$. Take charts $(U_{ij}, \varphi_{ij}), j = 1, 2, \dots$ such that $A_i \subseteq \bigcup_{j=1}^{\infty} U_{ij}$. Then $\{U_{ij}\}_{i,j}^{\infty}$ are countably many open sets that containing A , such that for each i, j ,

$$\varphi_{ij}(A \cap U_{ij}) \subseteq \bigcup_{i=1}^{\infty} \varphi_{ij}(A_i \cap U_{ij})$$

is a countable union of measure zero sets, thus itself is measure zero. Then it follows from Lemma 6.1.6 that A is of measure zero. $\square$

Proposition 6.1.8

Suppose M is a smooth manifold with or without boundary and $A \subseteq M$ has measure zero in M . Then $M \setminus A$ is dense in M .

Proof. If $M \setminus A$ is not dense. Then A contains a nonempty open subset of M , which implies that there is a chart (V, ψ) , such that $\psi(A \cap V)$ is a nonempty open subsets of $\mathbb{R}^n$. It is impossible that $\psi(A \cap V)$ has measure zero at the same time that it is open. $\square$

6.2 Sard's Theorem

Theorem 6.2.1 ▶ Sard's Theorem

Suppose M and N are smooth manifolds with or without boundary and $F : M \rightarrow N$ is a smooth map. Then the set of critical values of F has measure zero in N .

6.3 Tubular Neighborhoods

6.4 Transversality

Definition 6.4.1 ▶ Intersect Transversely

- Two embedded submanifolds $S, S' \subseteq M$ are said to **intersect transversely** if for each $p \in S \cap S'$, then tangent spaces $T_p S$ and $T_p S'$ together span $T_p M$.
- Let $F : N \rightarrow M$ be a smooth map and $S \subseteq M$ be an embedded submanifold. We say F is **transverse** to S , if for every $x \in F^{-1}(S)$, the spaces $T_{F(x)} S$ and $dF_x(T_x N)$ together span $T_{F(x)} M$.

Remark.

$S, S' \subseteq M$ are intersect transversely, if and only if the inclusion map $\iota : S \hookrightarrow M$ is transverse to S' .

Theorem 6.4.2

Suppose N and M are smooth manifolds and $S \subseteq M$ is an embedded submanifold.

- (a) If $F : N \rightarrow M$ is a smooth map that is transverse to S , then $F^{-1}(S)$ is an embedded submanifold of N whose codimension is equal to the codimension of S in M .
- (b) If $S' \subset M$ is an embedded submanifold that intersects S transversely, then $S \cap S'$ is an embedded submanifold of M whose codimension is equal to the sum of the codimensions of S and S' .

Corollary 6.4.3

Suppose N and M are smooth manifolds, $S \subseteq M$ is an embedded submanifold of codimension k , and $F : N \rightarrow M$ is a submersion. Then $F^{-1}(S)$ is an embedded codimension- k submanifold of N .

Proof. If F is a submersion, it is automatically transverse to every embedded submanifold $S \subseteq M$. □

Smooth Vector Fields

7.1 Vector Fields on Manifold

Definition 7.1.1

If M is a smooth manifold with or without boundary, a **vector field on M** is a section of the map $\pi : TM \rightarrow M$. More concretely, a vector field is a continuous map $X : M \rightarrow TM$, usually written $p \mapsto X_p$, with the property that

$$\pi \circ X = \text{Id}_M, \quad (7.1)$$

or equivalently, $X_p \in T_p M$ for each $p \in M$.

Remark.

- For a “vector field” that may be not continuous, we call it **rough vector field on M** .
- Under the canonical smooth structure of M and TM , we can define the smoothness of X .

Definition 7.1.2 ▶ component functions of a vector field

Suppose M is a smooth n -manifold (with or without boundary). If $X : M \rightarrow TM$ is a rough vector field and $(U, (x^i))$ is any smooth coordinate chart for M , we can write the value of X at any point $p \in U$ in terms of the coordinate basis vectors:

$$X_p = X^i(p) \frac{\partial}{\partial x^i} \Big|_p. \quad (7.2)$$

This defines n functions $X^i : U \rightarrow \mathbb{R}$, called the **component functions of X** in the given chart.

Proposition 7.1.3 ► Smoothness Criterion for Vector Fields

Let M be a smooth manifold with or without boundary, and let $X : M \rightarrow TM$ be a rough vector field. If $(U, (x^i))$ is any smooth coordinate chart on M , then the restriction of X to U is smooth if and only if its component functions with respect to this chart are smooth.

Proposition 7.1.4

Let M be a smooth manifold with or without boundary, and let $X : M \rightarrow TM$ be a rough vector field. The following are equivalent:

- (a) X is smooth.
- (b) For every $f \in C^\infty(M)$, the function Xf is smooth on M .
- (c) For every open subset $U \subseteq M$ and every $f \in C^\infty(U)$, the function Xf is smooth on U .

Definition 7.1.5 ► Vector Field as a Derivation

A smooth vector field $X \in \mathfrak{X}(M)$ defines a map

$$\begin{aligned} X : C^\infty(M) &\rightarrow C^\infty(M) \\ f &\mapsto Xf \end{aligned}$$

which is linear over $\mathbb{R}$ and following the product rule for vector fields

$$X(fg) = fXg + gXf$$

We call this map a **Derivation on $C^\infty(M)$** .

Proposition 7.1.6

Let M be a smooth manifold with or without boundary. A map $D : C^\infty(M) \rightarrow C^\infty(M)$ is a derivation if and only if it is of the form $Df = Xf$ for some smooth vector field $X \in \mathfrak{X}(M)$.

7.2 Lie Brackets

Definition 7.2.1 ► Lie Brackets

The operator $[X, Y] : C^\infty(M) \rightarrow C^\infty(M)$, called the *Lie bracket of X and Y* , defined by

$$[X, Y]f = XYf - YXf.$$

is a vector field.

Lie Groups

Integral Curves and Flows

9.1 Integral Curves

Definition 9.1.1 ► Integral Curve

If V is a vector field on M , an **integral curve of V** is a differentiable curve $\gamma : J \rightarrow M$ whose velocity at each point is equal to the value of V at that point:

$$\gamma'(t) = V_{\gamma(t)} \quad \text{for all } t \in J.$$

If $0 \in J$, the point $\gamma(0)$ is called the **starting point of γ** .

Finding integral curves boils down to solving a system of ordinary differential equations in a smooth chart. Suppose V is a smooth vector field on M and $\gamma : J \rightarrow M$ is a smooth curve. On a smooth coordinate domain $U \subseteq M$, we can write γ in local coordinates as

$$\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$$

Then the condition $\gamma'(t) = V_{\gamma(t)}$ for γ to be an integral curve of V can be written

$$\dot{\gamma}^i(t) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)} = V^i(\gamma(t)) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)},$$

which reduces to the following autonomous system of ordinary differential equations:

$$\begin{aligned} \dot{\gamma}^1(t) &= V^1(\gamma^1(t), \dots, \gamma^n(t)), \\ &\vdots \\ \dot{\gamma}^n(t) &= V^n(\gamma^1(t), \dots, \gamma^n(t)). \end{aligned}$$

Proposition 9.1.2

Let V be a smooth vector field on a smooth manifold M . For each point $p \in M$, there exists $\varepsilon > 0$ and a smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ that is an integral curve of V starting at p .

Proof. The ODE system corresponding to the condition

$$\gamma'(t) = V_{\gamma(t)}$$

is smooth. Thus it satisfies the local Lipschitz condition. Then the proposition follows from the Picard-Lindelöf Theorem. $\square$

9.2 Flows

Definition 9.2.1

We define a **global flow** on M (also called a **one-parameter group action**) to be a continuous left $\mathbb{R}$ -action on M ; that is, a continuous map $\theta : \mathbb{R} \times M \rightarrow M$ satisfying the following properties for all $s, t \in \mathbb{R}$ and $p \in M$:

$$\theta(t, \theta(s, p)) = \theta(t + s, p), \quad \theta(0, p) = p. \quad (9.1)$$

Given a global flow θ on M , we define two collections of maps as follows:

- For each $t \in \mathbb{R}$, define a continuous map $\theta_t : M \rightarrow M$ by

$$\theta_t(p) = \theta(t, p).$$

The defining properties (8.1) are equivalent to the **group laws**

$$\theta_t \circ \theta_s = \theta_{t+s}, \quad \theta_0 = \text{Id}_M. \quad (9.2)$$

As is the case for any continuous group action, each map $\theta_t : M \rightarrow M$ is a homeomorphism, and if the flow is smooth, θ_t is a diffeomorphism.

- For each $p \in M$, define a curve $\theta^{(p)} : \mathbb{R} \rightarrow M$ by

$$\theta^{(p)}(t) = \theta(t, p).$$

The image of this curve is the orbit of p under the group action.

Proposition 9.2.2

If $\theta : \mathbb{R} \times M \rightarrow M$ is a smooth global flow, for each $p \in M$ we define a tangent vector $V_p \in T_p M$ by

$$V_p = \theta^{(p)'}(0).$$

The assignment $p \mapsto V_p$ is a (rough) vector field on M , which is called the **infinitesimal generator of θ** .

The infinitesimal generator V of θ is a smooth vector field on M , and each curve $\theta^{(p)}$ is an integral curve of V .

9.2.1 Fundamental Theorem on Flows

Definition 9.2.3

- If M is a manifold, a **flow domain** for M is an open subset $\mathcal{D} \subseteq \mathbb{R} \times M$ with the property that for each $p \in M$, the set $\mathcal{D}^{(p)} = \{t \in \mathbb{R} : (t, p) \in \mathcal{D}\}$ is an open interval containing 0.
- A **flow** on M is a continuous map $\theta : \mathcal{D} \rightarrow M$, where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain, that satisfies the following group laws :

– For all $p \in M$,

$$\theta(0, p) = p, \quad (9.3)$$

– and for all $s \in \mathcal{D}^{(p)}$ and $t \in \mathcal{D}^{(\theta(s, p))}$ such that $s + t \in \mathcal{D}^{(p)}$,

$$\theta(t, \theta(s, p)) = \theta(t + s, p). \quad (9.4)$$

We sometimes call θ a **local flow** to distinguish it from a global flow as defined earlier. The unwieldy term **local one-parameter group action** is also used.

Note.

- **flows domain** 要求每一个空间节点上, 时间的变化范围都是包含 0 的开区间
- 一个 **flow** 就是 M 上活动在 $\mathcal{D}$ 内关于时间变化的一个局部左 $\mathbb{R}$ -作用.

Definition 9.2.4

If θ is a flow, we define $\theta_t(p) = \theta^{(p)}(t) = \theta(t, p)$ whenever $(t, p) \in \mathcal{D}$, just as for a global flow. For each $t \in \mathbb{R}$, we also define

$$M_t = \{p \in M : (t, p) \in \mathcal{D}\}, \quad (9.5)$$

so that

$$p \in M_t \Leftrightarrow t \in \mathcal{D}^{(p)} \Leftrightarrow (t, p) \in \mathcal{D}.$$

If θ is smooth, the **infinitesimal generator of θ** is defined by $V_p =$

$$\theta^{(p)'}(0).$$

Note.

- 当我们谈论点在 t 时间后的行为时, 只能对 M_t 上的点讨论.
- 无穷小生成元就是每一点处运动曲线的起始切向量.

Theorem 9.2.5 ► Fundamental Theorem on Flows

Let V be a smooth vector field on a smooth manifold M . There is a unique smooth maximal flow $\theta : \mathcal{D} \rightarrow M$ whose infinitesimal generator is V . This flow has the following properties:

- For each $p \in M$, the curve $\theta^{(p)} : \mathcal{D}^{(p)} \rightarrow M$ is the unique maximal integral curve of V starting at p .
- If $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\theta(s,p))}$ is the interval $\mathcal{D}^{(p)} - s = \{t - s : t \in \mathcal{D}^{(p)}\}$.
- For each $t \in \mathbb{R}$, the set M_t is open in M , and $\theta_t : M_t \rightarrow M_{-t}$ is a diffeomorphism with inverse θ_{-t} .

Note.

- 向量场 V 的最大流 $\theta : \mathcal{D} \subset \mathbb{R} \times M \rightarrow M$ 将从 M 中各点出发的极大积分曲线统一表示为 $\gamma_p(t) = \theta(t, p)$. 相对于单纯的一族积分曲线, 最大流 θ 还同时编码了解对初值 p 的光滑依赖, 即 θ 作为 (t, p) 的函数是光滑的
- 直观上, [(c)] 用平移来看. 划定一个区域, 将这些区域整体平移, 删去那些跑到区域外的点, 删去的点正好能补齐因为平移而产生的空缺, 这体现了平移关于时间的可逆性, 平移就是一种特殊的流.

9.2.2 Complete Vector Fields

9.3 Lie Derivatives

Definition 9.3.1

Suppose M is a smooth manifold, V is a smooth vector field on M , and θ is the flow of V . For any smooth vector field W on M , define a rough vector field on M , denoted by $\mathcal{L}_V W$ and called the **Lie derivative of W**

with respect to V , by

$$\begin{aligned} (\mathcal{L}_V W)_p &= \left. \frac{d}{dt} \right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) \\ &= \lim_{t \rightarrow 0} \frac{d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) - W_p}{t}, \end{aligned} \quad (9.6)$$

provided the derivative exists. For small $t \neq 0$, at least the difference quotient makes sense: θ_t is defined in a neighborhood of p , and θ_{-t} is the inverse of θ_t , so both $d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)})$ and W_p are elements of $T_p M$.

Theorem 9.3.2

If M is a smooth manifold and $V, W \in \mathfrak{X}(M)$, then $\mathcal{L}_V W = [V, W]$.

9.4 Commuting Vector Fields

Definition 9.4.1

If θ is a smooth flow, a vector field W is said to be **invariant under** θ if W is θ_t -related to itself for each t ; more precisely, this means that $W|_{M_t}$ is θ_t -related to $W|_{M_{-t}}$ for each t , or equivalently that $d(\theta_t)_p(W_p) = W_{\theta_t(p)}$ for all (t, p) in the domain of θ .

Theorem 9.4.2

For smooth vector fields V and W on a smooth manifold M , the following are equivalent:

- (a) V and W commute.
- (b) W is **invariant under the flow** of V .
- (c) V is **invariant under the flow** of W .

Theorem 9.4.3 ► Canonical Form for Commuting Vector Fields

Let M be a smooth n -manifold, and let $(V_1, \dots, V_k)$ be a linearly independent k -tuple of smooth commuting vector fields on an open subset $W \subseteq M$.

- For each $p \in W$, there exists a smooth coordinate chart $(U, (s^i))$

centered at p such that $V_i = \partial/\partial s^i$ for $i = 1, \dots, k$.

- If $S \subseteq W$ is an embedded codimension- k submanifold and p is a point of S such that $T_p S$ is complementary to the span of $(V_1|_p, \dots, V_k|_p)$, then the coordinates can also be chosen such that $S \cap U$ is the slice defined by $s^1 = \dots = s^k = 0$.

Exponential Map

Distributions and Foliations

11.1 Distributions and Involutivity

Definition 11.1.1

Let M be a smooth manifold.

- A **distribution on M of rank k** is a rank- k subbundle of TM . It is called a **smooth distribution** if it is a smooth subbundle.
- Often a rank- k distribution is described by specifying for each $p \in M$ a linear subspace $D_p \subseteq T_p M$ of dimension k , and letting $D = \bigcup_{p \in M} D_p$. It then follows from the local frame criterion for subbundles (Lemma 10.32) that D is a smooth distribution if and only if each point of M has a neighborhood U on which there are smooth vector fields $X_1, \dots, X_k : U \rightarrow TM$ such that $X_1|_q, \dots, X_k|_q$ form a basis for D_q at each $q \in U$. In this case, we say that D is the distribution (locally) **spanned by the vector fields $X_1, \dots, X_k$** .

11.1.1 Integral Manifolds and Involutivity

Definition 11.1.2

Suppose $D \subseteq TM$ is a smooth distribution. A nonempty immersed submanifold $N \subseteq M$ is called an **integral manifold of D** if $T_p N = D_p$ at each point $p \in N$.

Definition 11.1.3

Suppose D is a smooth distribution on M . We say that D is **involutive** if given any pair of smooth local sections of D ^a, their Lie bracket is also a local section of D .

^a(i.e., smooth vector fields X, Y defined on an open subset of M such that $X_p, Y_p \in D_p$ for each p)

11.2 The Frobenius Theorem

Definition 11.2.1

Given a rank- k distribution $D \subseteq TM$, let us say that a smooth coordinate chart (U, φ) on M is **flat for D** if $\varphi(U)$ is a cube in $\mathbb{R}^n$, and at points of U , D is spanned by the first k coordinate vector fields $\partial/\partial x^1, \dots, \partial/\partial x^k$.

Definition 11.2.2

We say that a distribution $D \subseteq TM$ is **completely integrable** if there exists a flat chart for D in a neighborhood of each point of M . Obviously, every completely

Theorem 11.2.3 ▶ Frobenius

Every **involutive distribution** is **completely integrable**.

11.3 Foliations

Definition 11.3.1

Let M be a smooth n -manifold, and let $\mathcal{F}$ be any collection of k -dimensional submanifolds of M .

- A smooth chart (U, φ) for M is said to be **flat for $\mathcal{F}$** if $\varphi(U)$ is a cube in $\mathbb{R}^n$, and each submanifold in $\mathcal{F}$ intersects U in either the empty set or a countable union of k -dimensional slices of the form $x^{k+1} = c^{k+1}, \dots, x^n = c^n$.
- We define a **foliation of dimension k on M** to be a collection $\mathcal{F}$ of disjoint, connected, nonempty, immersed k -dimensional submanifolds of M (called the **leaves of the foliation**), whose union is M , and such that in a neighborhood of each point $p \in M$ there exists a flat chart for $\mathcal{F}$.